

Erwan Scornet

Curriculum Vitae

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Research interests

My research work focuses on statistical learning theory, in particular non-parametric algorithms such as random forests. This class of algorithms exhibits good predictive performance that are not yet fully explained by theory. Therefore analyzing random forests from a theoretical perspective is both challenging and very exciting! I am also interested in the notion of explainable AI, i.e., providing insights about how algorithm predictions are built (variable importance, list of simple decision rules). I am also working on missing value theory, trying to design algorithms that can easily handle incomplete data and understand the impact of imputation on predictive performances. More recently, I started exploring the connections between random forests and neural networks, another kind of non-parametric algorithms with tremendous empirical performance. I also work on causal inference, studying the properties of causal measure in generalization settings (extending the finding from a clinical trial to a target population) and establishing asymptotic (or even non asymptotic) results on several estimators (G-formula, IPW, doubly-robust estimators...).

Keywords: Random forests, missing data, interpretability, causality, neural networks.

Training

- 2009-2013 **Degree from École Normale Supérieure de Paris** in *mathematics (with a minor in physics)*.
- 2009-2010 **Bachelor in mathematics and physics**, with honors (*bien/très bien*).
- 2010-2011 **Master thesis on Metropolis-Hasting algorithm on Riemannian varieties**. Several other research topics were *spherical black holes, modelling growing interface, introduction to statistical learning*.
- 2011-2012 **Master degree on Random modelling** from university Paris 7, with honors (*très bien*).
- 2012 **Internship at Robert Debré Hospital (Paris)** on *glycemic variability for diabetic patients* (6 months).
- 2012-2015 **PhD in statistics** on "*Learning with random forests*" supervised by Gérard Biau (LSTA - University Paris 6) et de Jean-Philippe Vert (Mines - Institut Curie)
Reviewers: Peter Bühlmann, Pierre Geurts.
Jury: Sylvain Arlot, Arnaud Guyader
- March 2016 **Academic visit with Professor Zaid Harchaoui to the Courant Institute (New-York)**.

- 2016 - 2023 **Assistant Professor in Applied Mathematics**, at *École Polytechnique in Paris* (CMAP).
- Dec. 2020 **Habilitation à diriger des recherches** *Random forests, interpretability, neural networks and missing values*.
Reviewers: Sylvain Arlot, Giles Hooker, Gabor Lugosi
Jury: Florence D'Alché-Buc, Olga Klopp, Eric Moulines.
- 2023 - ... **Lecturer in Applied Mathematics**, at *Sorbonne Université in Paris* (LPSM, SCAI).

Teaching

- 2012-2013 **Tutorial sessions in mathematics (TD, 24h)** for first-year bachelor students in *Paris Sciences et Lettres (PSL)*.
- 2013-2014 **Tutorial sessions in mathematics (TD, 24h)** for second-year bachelor students in *Paris Sciences et Lettres (PSL)*.
- 2013-2015 **Probability and Statistics (New course, 48h)** for second-year bachelor students in *economy*.
- 2016-2019 **Tutorial sessions in Machine Learning I (TD/TP, 18h)** for first-year graduate students in *"Data science for business" (X-HEC)*.
- 2016-2017 **Tutorial sessions in "The Art of Regression" (TD, 36h)** for third-year engineering students.
- 2016-2018 **Tutorial sessions in Machine Learning II (TD/TP, 18h)** for third-year engineering students.
- 2016-2018 **Tutorial sessions in "Statistics in Action" (TD/TP, 18h)** for third-year engineering students.
- 2017-2019 **Machine Learning Course (Course, 9h)** for first-year graduate students in *"Data science for business" (X-HEC)*.
- 2018-2023 **Deep Learning Course (New course, 21h)** for second-year master students.
- 2019-2020 **Tutorial sessions in Deep Learning (TP, 21h)** for second-year master students.
- 2019-2023 **Deep Learning and Optimization Course (lectures and lab sessions)** for first-year graduate students in *"Data science for business" (X-HEC)*.
- 2018-2023 **Learning Theory Course (New course, 12h)** for third year engineering students.
- 2022-2024 **MOOC Machine and Deep Learning, AI Certificate, IP Paris**.
- 2023-2026 **Data science Course and Tutorials (40h)** for non-specialized second-year bachelor students.
- 2023-2026 **Course and tutorial on tree-based methods and deep learning (22h)** for M2 students.
- 2023-2026 **Course and tutorial on machine learning and CNN (one week)** for non-specialized students (medicine).

Besides teaching courses, I gave oral examination training to Bachelor students for two years in some preparatory class for the "grandes écoles". From 2009 to 2014, I also participated in a monthly mentoring program named TalENS directed towards high school students from Paris suburb in order to encourage them to keep studying beyond their 'baccalauréat'. In particular, I co-organized a week

of intensive classes for high school students in August 2013 and August 2014.

Past responsibilities

- 2013, 2014 **Co-organizer of a week of intensive classes** for high school students taking place at École Normale Supérieure.
- 2013 - 2015 **Co-organizer of the PhD students seminar**, at LSTA, University Paris 6.
- 2016 - 2017 **Co-organizer of the "Café des statistiques" weekly seminar**, at École Polytechnique.
- 2017, 2018 **Co-organizer of the one-day conference Young Statistician and Probabilists**, at Institut Henri Poincaré.
- 2017, 2018 **Co-head of the third year of Applied Mathematics** at École Polytechnique.
- 2017, 2018 **Co-organizer of the Data Science Summer School (DS3)**, École Polytechnique.
- 2017 - 2019 **Member of the scientific committee of the DS3**, École Polytechnique.
- 2017 - 2023 **In charge of data science third-year projects**.
- 2017 - 2023 **Co-director of the Master (MScT) "Artificial Intelligence and Advanced Visual Computing"**, created in September 2018 at École Polytechnique.
- 2018 - 2023 **Co-director of the internships of first and second year in the "Artificial Intelligence and Advanced Visual Computing" training**.
- 2020 - 2023 **Scientific responsible of the Artificial intelligence program of Labex Mathématiques Hadamard (LMH)** with Frédéric Chazal.
- 2024 - 2025 **Member of the scientific committee of Salon de la culture mathématique** an annual event in Paris for high-school students.

Current responsibilities

- 2012 - **Reviewer** for the *Journal of Machine Learning Research*, *The Annals of Statistics*, *Electronic Journal of Statistics*, *International Conference on Machine Learning*, *Neural Information Processing Systems* and others.
- 2023 - ... **Co-director of the minor degree in Data Science and Artificial Intelligence** for second and third-year Bachelor students at Sorbonne University.
- 2026 - ... **Co-creator and managing board member of the PCI Statistics and Machine Learning** a role equivalent to journal editor.
- July 2026 **Co-organizer of a two-days Workshop at King's college** on decision tree methods.
- 2026 - 2027 **Member of the organization committee of EUROCIM 2027**.

Jury

- June 2019 **Jury member and reviewer of the PhD defense of Irving Gomez Mendez**, entitled "*Random forests and autoencoders with missing data*", CIMAT, Mexico.
- June 2019 **Jury member and reviewer of the PhD defense of Antonio Sutura**, "*Importance measures derived from random forests: characterisation and extension*", Liege University, Belgium.

- June 2021 **Member of the selection committee for an assistant professor position, Université Gustave Eiffel.**
- June 2022 **Member of the selection committee for an assistant professor position, Université de Bordeaux.**
- December 2022 **Jury member of the PhD defense of Baptiste Kerleguer, “Multi-fidelity surrogate modeling adapted to functional outputs for uncertainty quantification of complex models”, École polytechnique, France.**
- December 2022 **Jury member of the PhD defense of Clément Bénése, “On the links between Global Sensitivity Analysis and Algorithmic Fairness for eXplainable and Fair Machine Learning”, Toulouse III University, France.**
- February 2023 **Reviewer of the PhD defense of Jean-Samuel Leboeuf, “On the generalization properties of VC classes and application to decision trees”, Laval University, Canada.**
- February 2024 **Reviewer of the PhD defense of Florian Lalande, “Planetary Systems Insights through Numerical Data Imputation Algorithms and Machine Learning”, Okinawa Institute of Science and Technology, Japan.**
- June 2024 **President of the jury for the PhD Defence of Ahmad Chamma, “Statistical Interpretation of High-Dimensional Complex Prediction Models for Biomedical Data”, Inria Saclay, France.**
- July 2024 **Jury member of the PhD defense of Amin Dhaou, “Interpretable and Causal Analysis for Multivariate Time Series”, École polytechnique, France.**
- October 2024 **Reviewer of the PhD defense of Virgile Foy, “Apprentissage de représentations sur des surfaces 3D et analyse de nuages de points, application à la conception aéronautique assistée par ordinateur”, Toulouse III University, France.**
- November 2024 **Jury member of the PhD defense of Julien Demange-Chryst, “Nouvelles méthodes d’estimation par échantillonnage préférentiel pour l’analyse de sensibilité et l’estimation d’évènements rares”, Toulouse III University, France.**
- April 2025 **Jury member of Inria committee recruitment for CR positions, Paris center.**

Talks

- Dec. 2012 **PhD students seminar, LSTA, Paris 6.**
- March 2013 **Computational Biology team seminar, Institut Curie.**
- June 2013 **Journées de statistique, SFDS, Toulouse.**
- Dec. 2013 **NIPS, workshop MLCB, Reno, Nevada.**
- March 2014 **PhD students seminar, MAP5, Paris 5.**
- March 2014 **Poster at the colloquium "Digital: Big scale and complexity", IMT.**
- April 2014 **"Random modelling and applications" seminar, Caen.**
- April 2014 **SMILE seminar, ENS Ulm.**
- June 2014 **Journées de statistique, SFDS, Rennes.**
- January 2015 **Maths & companies week, organized by AMIES, Paris.**
- January 2015 **Young Statisticians and Probabilists conference, Institut Henri Poincaré.**

Feb. 2015 **Probability and statistics seminar**, *Institut de Mathématiques et de Modélisation de Montpellier*.

June 2015 **Journées de statistique, SFDS**, *Lille*.

Sept. 2015 **MODAL seminar**, *INRIA Lille*.

October 2015 **Statistic seminar**, *Université de Strasbourg*.

Nov. 2015 **STA seminar**, *Telecom ParisTech*.

January 2016 **Statistic seminar**, *Université de Toulouse*.

June 2016 **Journées de statistique, SFDS**, *Montpellier*.

August 2016 **MAS conference**, *Grenoble*.

Sept. 2016 **Statistic seminar**, *Compiègne*.

January 2017 **Statistic seminar**, *Agro ParisTech*.

May 2017 **Statistic seminar**, *MAP5*.

May 2017 **Statistic seminar**, *École des ponts*.

July 2017 **Joint Statistical Meetings (JSM 2017)**, *Baltimore*.

January 2018 **Invited speaker at a workshop on the Interface of Machine Learning and Statistical Inference**, *Banff, Canada*.

March 2018 **Conference to prepare high school students to a conference by Yann LeCun**, *Bourg La Reine*.

April 2018 **Statistic seminar**, *Rennes*.

April 2018 **Statistic seminar**, *Versailles*.

May 2018 **Statistic seminar**, *Telecom ParisTech*.

January 2019 **Probability and Statistic seminar**, *Lille university*.

January 2019 **Invited speaker at Journée Statistique / Apprentissage à Paris Saclay**, *IHES*.

March 2019 **Invited speaker at BNP Cardiff conference**, *Nanterre*.

March 2019 **Invited speaker at Sanofi conference**, *Lyon*.

May 2019 **Statistic seminar**, *Paris 7 university*.

October 2019 **Invited Speaker at Conference on Big Data and Machine Learning in Econometrics, Finance, and Statistics**, *Chicago university*.

October 2019 **Invited speaker in a thematic class**, *Puebla, Mexico*.

January 2021 **Statistic seminar**, *Angers*.

March 2021 **Statistic seminar for M2 students**, *University Paris-Saclay*.

May 2021 **Statistical workshop**, *Amsterdam School of Economics*.

July 2021 **ENBIS Workshop: Interpretability for Industry 4.0**, *Naples*.

January 2022 **Statistical seminar for M2 students**, *Ecole Polytechnique*.

Feb. 2022 **Statistical seminar**, *Berlin, Weierstrass Institute*.

Feb. 2022 **Statistical seminar**, *Orsay*.

March 2022 **Statistical seminar**, *Vannes*.

March 2022 **Statistical seminar**, *Montpellier*.

April 2022 **Statistical seminar**, *Centre Borelli, ENS Paris-Saclay*.

- June 2022 **Journées de statistique, SFDS, Lyon.**
- Sept. 2022 **Statistical and Optimization seminar, Toulouse.**
- Nov. 2022 **Econometrics seminar, Cambridge.**
- Dec. 2022 **International Conference on Statistics and Data Science, Florence.**
- Feb. 2023 **Econometrics seminar, Amsterdam University.**
- Feb. 2023 **Data Science seminar, London School of Economics.**
- April 2023 **Stat'Learn conference, Montpellier.**
- June 2023 **Quarter on Causality, conference, Institut Pascal, Paris Saclay.**
- Dec. 2023 **ICSDS Conference, Lisbonne.**
- January 2024 **Round table on AI challenges, Artefact research center opening, Paris.**
- May 2024 **SIERRA Team, INRIA Paris.**
- June 2024 **Workshop "Mathematical foundation of AI", organized by SCAI, Sorbonne University Paris.**
- Dec. 2024 **SAMM seminar, University Paris 1.**
- Dec. 2024 **Seminar of students in statistics, Orsay Paris.**
- January 2024 **MAP5 seminar, University Paris Cité Paris.**
- Feb. 2025 **Statistics and Optimisation seminar, IMT Toulouse.**
- Feb. 2025 **PreMedical seminar, Inria Montpellier.**
- March 2025 **Statistical Parisian seminar, IHP Paris.**
- April 2025 **Statistics and Data Science conference Milan.**
- April 2025 **Bootcamp Deep Learning and Spatial Data SCAI Sorbonne Université.**
- October 2025 **A 5-days workshop on missing values Evian les bains website.**
- October 2025 **Statistical seminar Vannes.**
- October 2025 **A four-day workshop on causality and explainability SCAI website.**
- Nov. 2025 **Statistical seminar Gustave Eiffel University Paris.**
- Dec. 2025 **ICSDS conference Sevilla Spain.**
- January 2026 **Statistical seminar Warwick UK.**
- Feb. 2026 **Statistical seminar Versailles.**
- April 2026 **Bootcamp Deep Learning and Spatial Data SCAI Sorbonne Université.**
- April 2026 **Bootcamp Deep Learning and Spatial Data SCAI Sorbonne Université.**
- June 2026 **Statistical Parisian seminar, IHP Paris.**
- July 2026 **Conference IFCS Milan website.**

--- Grant and Awards

- April 2016 **Grant from the Pierre Ledoux Foundation to fund a visit to the Courant Institute, New-York.**
- August 2016 **Recipient of the Jacques Neveu 2016 Prize, rewarding a PhD thesis in probability or statistics, received during the MAS conference, Grenoble.**
- June 2018 **Missing DataIA project funded by DataIA institute.**

- June 2020 **PhD thesis funded by SCAI (Sorbonne Center for Artificial Intelligence) on neural networks.**
- June 2021 **Recipient of an Emergence Project, provided by Paris city to develop a topic in machine learning (150.000 euros).**
- June 2025 **PhD thesis funded by CLuster AI on the relation between fairness and missing data.**

Supervision - Internships

- July 2013 **Co-supervision of the internship of Nelly Alandou, Clément Benesse et Pablo Le Henaff, first-year students in Paris Sciences et Lettres.**
- Spring 2014 **Supervision of the internship of Arthur Pajot, third-year student at University Paris 6.**
- Avril 2015 **Supervision of the internship of Charlotte Rougier, lycéenne en première S.**
- 2016 **Supervision of the internship of Jurriaan Parie, third-year student at University Paris 6.**
- 2016 - **Supervision of several third-year projects in data science, at École Polytechnique.**
- 2020 **Co-supervision of the M2 internship of Ludovic Arnould with Claire Boyer (LPSM).**
- 2021 **Co-supervision of the post-doctoral research of Marine Le Morvan with Julie Josse (INRIA Montpellier) and Gaël Varoquaux (INRIA Saclay).**
- 2021 **Co-supervision of the M2 internship of Alexis Ayme with Claire Boyer (LPSM) and Aymeric Dieuleveut (CMAP).**
- 2022 **Co-supervision of the M2 internship of Patrick Lutz with Claire Boyer (LPSM).**
- 2022 **Co-supervision of the M2 internship of Khadim Sene with Alexandre Py-Renaudie (IPP, IPVF) and Jean-François Guillemoles (IPP, IPVF).**

Supervision - PhD

- 2016 - 2020 **Co-supervision of the PhD thesis of Jaouad Mourtada entitled Contributions to statistical learning: density estimation, expert aggregation and random forests with Stéphane Gaïffas (University Paris 7).**
- 2018 - 2019 **Co-supervision of the PhD thesis of Nicolas Prost with Julie Josse (Ecole Polytechnique) and Gael Varoquaux (Inria).**
- 2018 - 2021 **Co-supervision of the PhD thesis (CIFRE) of Clement Benard entitled Forêts aléatoires et interprétabilité des algorithmes d'apprentissage with Gérard Biau (LPSM) and Sebastien Da Veiga (Safran).**
- 2020 - 2023 **Co-supervision of the PhD thesis of Ludovic Arnould with Claire Boyer (LPSM).**
- 2020 - 2023 **Co-supervision of the PhD thesis of Bénédicte Colnet with Julie Josse (INRIA Montpellier) and Gaël Varoquaux (INRIA Saclay).**
- 2021 - 2024 **Co-supervision of the PhD thesis of Alexis Ayme with Claire Boyer (LPSM) and Aymeric Dieuleveut (CMAP).**

- 2023 - 2026 **Co-supervision of the PhD thesis of Ahmed Boughdiri** *with Julie Josse (Inria Montpellier).*
- 2024 - 2026 **Co-supervision of the Cifre PhD thesis of Abdoulaye Sakho** *with Emmanuel Malherbe (Artefact)).*
- 2025 - ... **Co-supervision of the Cifre (Withings) PhD thesis of Agathe Chabassier** *with Julie Josse (Inria Montpellier).*
- 2026 - ... **Co-supervision of the PhD thesis of Florian Soonaye** *with Christophe Denis (Paris 1) and Julie Josse (Inria Montpellier).*
- 2025 - 2026 **Co-supervision of the PostDoc Simon Roburin** *with Rafaël Pinot (Sorbonne Université).*

Book

- 2022 **Interpretability for Industry 4.0 : Statistical and Machine Learning Approaches**, *B. Iooss, R. Kenett, P. Secchi, B.M. Colosimo, F. Centofanti, C. Bénard, S. Da Veiga, E. Scornet, S. N. Wood, Y. Goude, M. Fasiolo* Editors: A. Lepore, B. Palumbo, J.-M. Poggi, Springer.

Published papers

- 2015 **Consistency of random forests**, *E. Scornet, G. Biau, and J.-P. Vert* The Annals of Statistics, Vol. 43, pp. 1716-1741.
- 2016 **On the asymptotics of random forests**, *E. Scornet* Journal of Multivariate Analysis, Vol. 146, pp. 72-83.
- 2016 **Random forests and kernel methods**, *E. Scornet* IEEE Transactions on Information Theory, Vol. 62, pp. 1485-1500.
- 2016 **A Random Forest Guided Tour**, *G. Biau and E. Scornet* TEST, Vol. 25, pp. 197-227, with Discussion.
- 2016 **Promenade en forêts aléatoires**, *E. Scornet* MATAPLI, Vol. 111.
- 2017 **Kernel multitask regression for toxicogenetics**, *E. Bernard, Y. Jiao, E. Scornet, V. Stoven, T. Walter and J.-P. Vert* Molecular Informatics, Vol. 36.
- 2017 **Universal consistency and minimax rates for online Mondrian Forest**, *J. Mourtada, S. Gaïffas, E. Scornet* NIPS.
- 2017 **Tuning parameters in random forests**, *E. Scornet* ESAIM Procs, Vol. 60 pp. 144-162.
- 2018 **Impact of subsampling and tree depth on random forests**, *R. Duroux, E. Scornet* ESAIM: Probability and Statistics, Vol. 22, pp. 96-128.
- 2018 **Neural Random Forests**, *G. Biau, E. Scornet, J. Welbl*, Sankhya A, 1-40.
- 2020 **Minimax optimal rates for Mondrian trees and forests**, *J. Mourtada, S. Gaïffas, E. Scornet*, The Annals of Statistics, 48(4), 2253-2276.
- 2020 **Linear predictor on linearly-generated data with missing values: non consistency and solutions**, *M. Le Morvan, N. Prost, J. Josse, E. Scornet., G. Varoquaux*, AISTAT.

- 2020 **Neumann networks: differential programming for supervised learning with missing values**, *M. Le Morvan, J. Josse, T. Moreau, E. Scornet, G. Varoquaux* Oral, NeurIPS.
- 2021 **SIRUS: Stable and Interpretable RULE Set for Classification**, *C. B  nard, G. Biau, S. Da Veiga, E. Scornet*, Electronic Journal of Statistics 2021, Vol. 15, pp. 427-505.
- 2021 **Interpretable Random Forests via Rule Extraction**, *C. B  nard, G. Biau, S. Da Veiga, E. Scornet*, AISTAT.
- 2021 **AMF: Aggregated Mondrian Forests for Online Learning**, *J. Mourtada, S. Gaiffas, E. Scornet*, Journal of the Royal Statistical Society: Series B (Statistical Methodology), 83(3), 505-533.
- 2021 **Analyzing the tree-layer structure of Deep Forests**, *L. Arnould, C. Boyer, E. Scornet*, ICML.
- 2021 **What's a good imputation to predict with missing values?** *M. Le Morvan, J. Josse, E. Scornet, G. Varoquaux* Oral, NeurIPS.
- 2021 **Trees, forests, and impurity-based variable importance**, *E. Scornet*, Annales de l'Institut Henri Poincar  .
- 2022 **SHAFF: Fast and consistent SHAPley eFFect estimates via random Forests**, *C. B  nard, G. Biau, S. Da Veiga, E. Scornet*, AISTAT.
- 2022 **MDA for random forests: inconsistency, and a practical solution via the Sobol-MDA**, *C. B  nard, S. Da Veiga, E. Scornet*, Biometrika.
- 2022 **Near-optimal rate of consistency for linear models with missing values**, *A. Ayme, C. Boyer, A. Dieuleveut, E. Scornet* ICML.
- 2022 **Generalizing a causal effect: sensitivity analysis and missing covariates**, *B. Colnet, J. Josse, E. Scornet, G. Varoquaux* Journal of Causal Inference.
- 2023 **Is interpolation benign for random forests?**, *L. Arnould, C. Boyer, E. Scornet* AISTAT.
- 2023 **Sparse tree-based initialization for neural networks**, *P. Lutz, L. Arnould, C. Boyer, E. Scornet*. ICLR.
- 2023 **Naive imputation implicitly regularizes high-dimensional linear models**, *A. Ayme, C. Boyer, A. Dieuleveut, E. Scornet*. ICML.
- 2024 **(first submission in 2019) On the consistency of supervised learning with missing values**, *J. Josse, J.M. Chen, N. Prost, E. Scornet, G. Varoquaux* Statistical Papers.
- 2024 **Reweighting the RCT for generalization: finite sample error and variable selection**, *B. Colnet, J. Josse, G. Varoquaux, E. Scornet* JRSS-A.
- 2024 **Random features models: a way to study the success of naive imputation**, *A. Ayme, C. Boyer, A. Dieuleveut, E. Scornet*. ICML.
- 2025 **A primer on linear classification with missing data**, *A.D. Reyro Lobo, A. Ayme, C. Boyer, E. Scornet* AISTAT.
- 2025 **Quantifying Treatment Effects: Estimating Risk Ratios via Observational Studies**, *A. Boughdiri, J. Josse, E. Scornet* ICML.

- 2025 **Harnessing Mixed Features for Imbalance Data Oversampling: Application to Bank Customers Scoring**, A. Sakho, E. Malherbe, C.-E. Gauthier, E. Scornet ECML-PKDD, Applied Data Science track.
- 2025 **A Unified Framework for the Transportability of Population-Level Causal Measures**, A. Boughdiri, C. Berenfeld, J. Josse, E. Scornet NeurIPS.
- 2026 **Do we need rebalancing strategies? A theoretical and empirical study around SMOTE and its variants.**, A. Sakho, E. Malherbe, E. Scornet AISTAT.
- 2026 **Theory of Random Forests: A Review**, E. Scornet, G. Hooker Annual Review of Statistics and Its Application.
- 2026 **Real-world body composition and blood pressure changes following GLP-1 Receptor Agonist initiation: a causal inference study of connected device data**, A. Chabassier, P. Binon, J. Josse, E. Scornet, B. Vittrant Mayo Clinic Proceedings: Digital Health.

Submitted papers

- 2023 **Risk ratio, odds ratio, risk difference... Which causal measure is easier to generalize?**, B. Colnet, J. Josse, G. Varoquaux, E. Scornet.
- 2024 **What Is a Good Imputation Under MAR Missingness?**, J. Naf, J. Josse, E. Scornet.
- 2025 **Causal Meta-Analysis: Rethinking the Foundations of Evidence-Based Medicine**, C. Berenfeld, A. Boughdiri, B. Colnet, W. A. C. van Amsterdam, A. Bellet, R. Khellaf, E. Scornet, J. Josse.
- 2025 **Asymptotic Normality of Infinite Centered Random Forests - Application to Imbalanced Classification**, M. Mayala, E. Scornet, C. Tillier, O. Wintenberger.
- 2025 **When Pattern-by-Pattern Works: Theoretical and Empirical Insights for Logistic Models with Missing Values**, C. Muller, J. Josse, E. Scornet.
- 2025 **How to rank imputation methods?** J. Näf, K. Grzesiak, E. Scornet.
- 2026 **Principled Federated Random Forests for Heterogeneous Data** R. Khellaf, A. Bellet, J. Josse, E. Scornet.
- 2026 **On the Inherent Privacy Amplification of Missing Data** S. Roburin, R. Pinot, E. Scornet.
- 2026 **Increasing Missingness to Reduce Bias: Richardson-SGD with Missing Data** F. Genans, E. Scornet.

Hobbies

- From 2014 to 2019, Stage and lighting director of the musicals *Bloody Monday*, *Doré Mirador* and *Au bonheur des âmes*, original creation and production with a troupe of 40 artists.
- From 2018, amateur actor in the theater company *Dans de beaux drames* in *Le repas des fauves* (2019) and *Le porteur d'histoire* (2021).
- Chess, Tennis, Badminton, application of probability theory to understand board games ("Mon premier vergé").